EgoHAnG: Graph-Enhanced Horizon Aware Egocentric Action Anticipation

Pawanesh Kumar Vishwakarma[1[0009-0009-3492-9778]], Ananda S. Chowdhury[2[0000-0002-5799-3467]], and Abhimanyu Sahu[1*[0000-0001-8832-1842]]

[1] Department of Computer Science & Engineering, Motilal Nehru National Institute of Technology Allahabad, Prayagraj, India
`{pawanesh.2023rcs04, abhimanyus}@mnnit.ac.in`

[2] Department of Electronics & Telecommunication Engineering, Jadavpur University Kolkata, India.
`as.chowdhury@jadavpuruniversity.in`

Abstract. Egocentric action anticipation has become an increasingly popular topic in computer vision. Most existing state-of-the-art action anticipation approaches rely on multi-modal feature fusion and predominantly employ architectures such as 3D CNNs, LSTM, RULSTM, I3D, X3D-XS, lightweight models, and more recently, Transformer-based methods. However, most existing solutions rely primarily on sequential modeling, which often struggles to capture complex inter-frame relationships and fails to effectively handle short, mid, and long term cues appearing at different times in future. Graph-theoretic formulations, which are well suited for explicitly modeling such relationships, remain largely unexplored for this problem. In this paper, we propose EgoHAnG (**Ego**centric **H**orizon Aware Action **An**ticipation using **G**raphs), an end-to-end framework that integrates multimodal fusion with graph-based relational reasoning and transformer-based temporal modeling for egocentric action anticipation. Our approach first performs an early fusion of RGB and optical flow features using a pretrained ResNet-50 to capture complementary appearance and motion cues. We then construct a dynamic k-NN graph over the observed frames and apply graph based attention to model inter-frame relational structures, enabling robust capture of long-range temporal dependencies in egocentric videos. Finally, we introduce a horizon-aware decoder with learnable horizon queries to independently predict verb, noun, and action labels across multiple anticipation horizons, improving both short- and long-term action anticipation. Experimental evaluations on two standard benchmarks, EPIC-Kitchens and EGTEA Gaze+, demonstrate that the proposed method improves top-5 action recall by 1.1% and 3.73%, respectively, over current state-of-the-art methods. Our implementation and code are publicly available at `https://github.com/pawanesh-mnnit/EgoHANG`.

Keywords: Action Anticipation · Egocentric Video · Graph Enhanced Temporal Reasoning · Horizon Aware Transformer

1 Introduction

Egocentric devices equipped with cameras, such as GoPro and smart glasses, have gained increasing popularity because they capture the environment from the user's point of view and enable natural and effective support [4]. In egocentric vision, several tasks have been studied, including video summarization and action recognition [20,12]. Among them, action anticipation predicting the next action of the camera wearer from an observed video segment has received significant attention [7]. In this paper, the focus of this work is egocentric action anticipation. As defined in [4] and illustrated in Fig. 1, the task aims to recognize an upcoming action starting at time $T_{\mathbb{S}}$ by observing a video segment that precedes the action. Specifically, the observed segment starts at time $T_{\mathbb{S}}-(T_{obs}+T_{\alpha})$ and ends at time $T_{\mathbb{S}} - T_{\alpha}$, where the observation time T_{obs} represents the duration of the observed video segment, and the anticipation time T_{α} indicates how far in advance the action needs to be anticipated. We observe that egocentric action anticipation methods must effectively address two key sub-tasks: (1) summarizing the past observations to capture the context of the ongoing activity (e.g., "cut celery" in the observed segment shown in Fig. 1), and (2) formulating hypotheses about future actions, such as "wash celery", "cut celery", or "put celery in pan", which may occur in the unobserved future segment, as illustrated in Fig. 1.

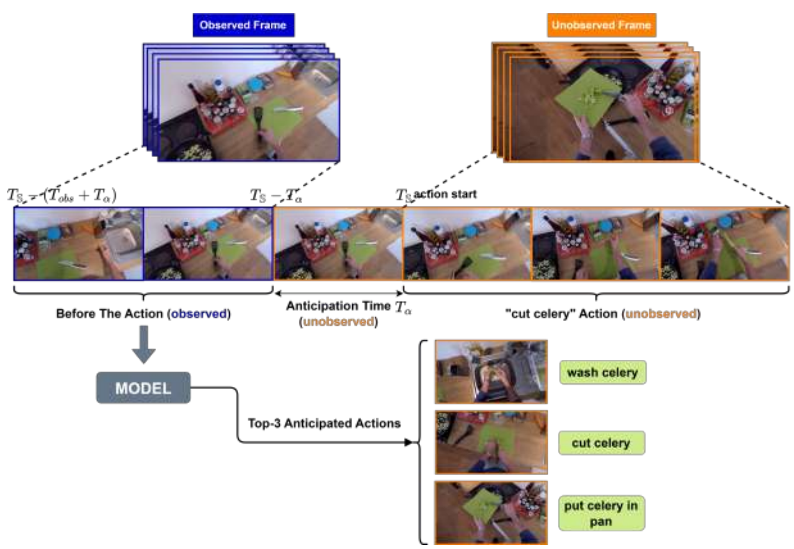

Fig. 1: Illustration of egocentric action anticipation. The model observes a sequence of past frames (blue, left) before the action occurs and predicts the future action during an unobserved anticipation interval (orange, right).

Most existing state-of-the-art action anticipation approaches rely on multimodal feature fusion and predominantly employ architectures such as 3D CNNs [2], LSTM, RULSTM [7], I3D and X3D-XS [6], lightweight models, and more recently, Transformer-based methods. However, existing solutions largely focus on sequential modeling and do not explore graph-theoretic formulations for this problem. Moreover, many methods rely solely on temporal encoders without explicitly modeling semantic relationships between frames, restricting their ability to reason over long anticipation horizons and do not explore graph-theoretic formulations for this problem.

In this work, we propose EgoHAnG, an end-to-end framework for egocentric action anticipation that integrates graph-based temporal reasoning with Transformer-based multimodal fusion. The framework performs early fusion of RGB and optical flow features extracted per frame using a pretrained ResNet-50, followed by PCA-based dimensionality reduction to obtain compact multimodal representations. We introduce a Graph-Enhanced Temporal Reasoning (GETR) module that jointly captures inter-frame relationships and global temporal dynamics by combining similarity-based graph attention with a Transformer encoder. The graph branch enables information exchange between semantically related frames, including non-adjacent ones, while the Transformer branch models long-range temporal dependencies. Their outputs are fused into a unified temporal representation. Finally, we design a Horizon-Aware Transformer Decoder (HATD) with learnable, horizon-specific queries to predict future actions at multiple anticipation times. Short horizons emphasize recent motion cues, whereas longer horizons exploit broader temporal context. The decoder generates separate predictions for verbs, nouns, and actions, enabling effective multi-horizon, time-aware egocentric action anticipation. Now, we summarize the main contributions of this work as follows.

1. We introduce an effective **feature extraction and observation window construction strategy**, where in each frame RGB and optical flow features are extracted using a pretrained ResNet-50, compressed using PCA and aligned with action annotations to form observation windows, while supervision is defined using time-based anticipation horizons.
2. We propose a **Graph-Enhanced Temporal Reasoning (GETR)** module that captures inter-frame relationships through similarity-based graph attention while modeling global temporal patterns with a Transformer encoder.
3. We design a **Horizon-Aware Transformer Decoder (HATD)** that uses learnable horizon queries to perform time-aware anticipation and generate separate predictions for verbs, nouns, and actions across multiple anticipation horizons.

2 Related Work

Action anticipation has evolved from third-person to egocentric settings. We review prior work in two areas on egocentric videos: Action Anticipation based on Single Modality and Action Anticipation based on Multimodal Modalities.

Action Anticipation based on Single Modality focuses on predicting the upcoming actions from the past and present video segments observed. Although early studies mainly explored this task in third-person videos [8,11], recent advances in egocentric vision [12,23] and the release of dedicated benchmark datasets [13,4] have significantly increased interest in first-person action anticipation. Early methods used temporal CNNs, LSTMs [7], and encoder–decoder [1] models to capture motion and temporal dynamics, hierarchical strategies improving predictions as more context became available; however, they struggle to model long-term dependencies. Recent work uses transformer models for egocentric action anticipation because they can capture long-term temporal information. These models perform well under fast and complex camera motion.

Action Anticipation based on Multimodal Modalities: In prior egocentric vision research, commonly used modalities include RGB, object cues, and optical flow [7,10]. Fusion is usually done by feature concatenation or late fusion of modality-specific models [4,17]. Although multimodal input improves robustness, most approaches [21] treat each modality separately and combine them only at later stages, limiting early cross-modal interaction. Consequently, these approaches struggle to capture the joint evolution of appearance and motion and lack explicit modeling of inter-frame relationships [19,16]. However, standard transformer-based models such as AVT [10], InAViT [19] and RAFTformer-2B [9] do not clearly model relationships between frames or filter out less useful observations. As a result, they may miss repeated object interactions or similar hand movements [7,23]. Some extensions add memory or extra attention, but explicit relational modeling and effective multimodal fusion are still limited.

To address these limitations, we adopt early fusion of RGB and Optical flow to jointly model appearance and motion cues. We further introduce a graph-based [3] reasoning module to explicitly capture inter-frame relationships and a horizon-aware transformer decoder to anticipate actions at multiple future times. By integrating relational learning with global temporal modeling, our framework addresses the weaknesses of late fusion strategies and purely temporal transformer-based approaches, leading to more robust egocentric action anticipation.

3 Proposed Methodology

Our solution pipeline consists of four steps: 1) Feature Extraction and Observation Window Construction, 2) Graph-Enhanced Temporal Reasoning (GETR), 3) Transformer-based Temporal Encoding, and 4) Horizon-Aware Transformer Decoder (HATD). Fig. 2 shows an overview, and each component is described in detail below.

3.1 Feature Extraction and Observation Window Construction

Initially, our framework selects RGB for appearance, object cues, and optical flow for motion dynamics, thereby capturing complementary information from the dataset. A video X consists of N frames, $X = \{x_1, x_2, \cdots, x_N\}$, with each frame containing features from the chosen modalities. Each frame is processed

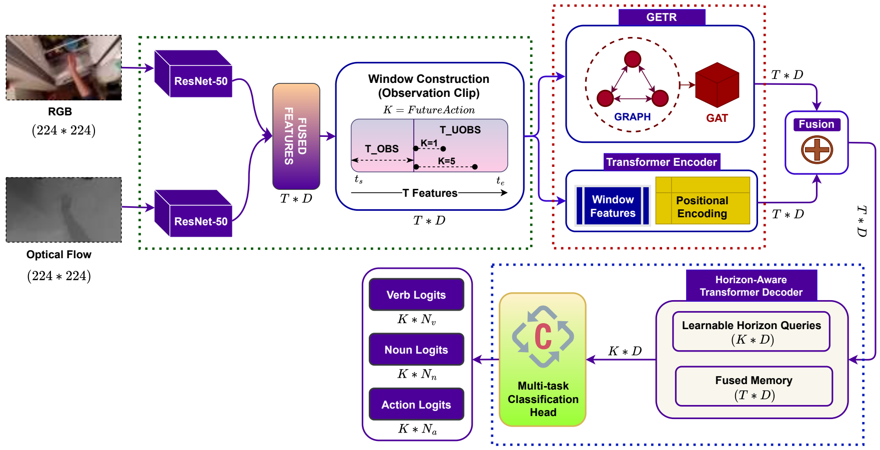

Fig. 2: Overview of the Proposed Solution: (1) Feature Extraction and Observation Window Construction (Green dotted box), (2) Graph-Enhanced Temporal Reasoning (GETR) and Temporal Encoding via Transformer Encoder (Red dotted box), and (3) Horizon-Aware Transformer Decoder (HATD) and Horizon-Specific Classification Heads (Blue dotted box).

individually by a shared ResNet-50 pretrained on ImageNet. The network's convolutional blocks, followed by global average pooling (GAP), produces 2048-dimensional feature vectors. ResNet-50 is chosen for its strong representation and good balance between accuracy and efficiency. This can be expressed as:

$$\mathbf{r}_i = f(\mathbf{x}_i^{rgb}) \in \mathbb{R}^{2048}, \qquad \mathbf{o}_i = f(\mathbf{x}_i^{flow}) \in \mathbb{R}^{2048}, \quad i = 1, 2, \ldots, N \tag{1}$$

where, $\mathbf{x}_i^{rgb}$ and $\mathbf{x}_i^{flow}$ are the RGB and optical flow inputs of the i-th frame, $\mathbf{r}_i$ and $\mathbf{o}_i$ are the corresponding 2048-dimensional feature vectors, and N is the total number of frames in the video. We use Principal Component Analysis (PCA) on RGB and optical flow features to reduce their sizes from 2048-D to 512-D dimensions. PCA keeps the most important information while greatly lowering computation and memory requirements. This can be represented using the following equations:

$$\mathbf{r}'_t = \mathcal{P}(\mathbf{r}_t), \qquad \mathbf{o}'_t = \mathcal{P}(\mathbf{o}_t), \tag{2}$$

where, $\mathcal{P}(\cdot)$ denotes the PCA projection operator and $\mathbf{r}'_t$, $\mathbf{o}'_t \in \mathbb{R}^D$. Since RGB captures appearance and optical flow captures motion, we combine them early at the feature level using a weighted fusion. In mathematical form, this is expressed as:

$$\mathbf{f}_t = \alpha\, \mathbf{r}'_t + (1-\alpha)\, \mathbf{o}'_t, \tag{3}$$

where, $\alpha \in [0, 1]$ controls the relative contributions of appearance and motion information. This produces a single, compact representation for each frame. By stacking these fused features across all observed frames, we get the final sequence, *i.e.*, $F = [\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_N] \in \mathbb{R}^{N \times D}$, which is used as an input for the observation window and the graph-based temporal reasoning modules. For each annotated action, the model observes only the first T_{OBS} frames of the fused feature sequence (see Fig. 2). Let t_s denote the starting frame index of an annotated action, and the corresponding fused features be $\{\mathbf{f}_{t_s}, \mathbf{f}_{t_s+1}, \ldots \mathbf{f}_{t_e}\}$. The observation window is defined as $F_{\text{obs}} = [\mathbf{f}_{t_s}, \mathbf{f}_{t_s+1}, \ldots, \mathbf{f}_{t_s+T_{\text{OBS}}-1}] \in \mathbb{R}^{T_{\text{OBS}} \times D}$. The dataset also provides semantic labels for the next $\mathbb{K}$ future actions. For each anticipation horizon $k \in \{1, \ldots, \mathbb{K}\}$, the ground truth includes the corresponding verb, noun, and combined action labels, denoted as:

$$\{Y_k^{verb}, Y_k^{noun}, Y_k^{action}\}_{k=1}^{\mathbb{K}}, \tag{4}$$

where $Y_k^{\text{verb}} = v_k$, $Y_k^{\text{noun}} = n_k$, and $Y_k^{\text{action}} = a_k$ respectively denote the ground-truth verb, noun, and composite action labels at anticipation horizon k, respectively. All frames beyond the observed window constitute the unobserved future region of length $T_{\text{UNOBS}} = T_{\text{e}} - T_{\text{OBS}}$, which the model must anticipate using only the multimodal descriptors in F_{obs}.

3.2 Graph-Enhanced Temporal Reasoning (GETR)

The fused observation clip $F_{\text{obs}} \in \mathbb{R}^{T_{\text{OBS}} \times D}$ contains rich multimodal cues, but egocentric videos often show rapid motion changes and complex frame dependencies. To address this, the GETR module jointly captures inter-frame relationships and global temporal context by combining similarity-based graph reasoning with Transformer-based temporal encoding. We build a feature similarity graph over the observed frames, where each frame is a node and edges are defined using cosine similarity between features $\mathbf{f}_i$ and $\mathbf{f}_j$ is computed as:

$$\text{sim}(i, j) = \frac{\mathbf{f}_i^\top \mathbf{f}_j}{\|\mathbf{f}_i\|_2 \, \|\mathbf{f}_j\|_2}. \tag{5}$$

For each node i, the k most similar nodes $\mathcal{N}_k(i)$ are selected and directed edges are constructed as $E = \{(i, j) \mid j \in \mathcal{N}_k(i)\}$. This graph structure is constructed once using the top-k similarity criterion and kept fixed across all GAT layers, while the node representations and attention weights are updated at each layer. Since the graph is based only on feature similarity, it does not encode temporal order. To propagate and refine information across these relations, we employ a multi-layer Graph Attention Network (GAT) [22]. At layer ℓ, the updated feature for node i is given by

$$\mathbf{h}_i^{(\ell)} = \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(\ell)} \, W^{(\ell)} \mathbf{h}_j^{(\ell-1)}\right), \tag{6}$$

where, $W^{(\ell)}$ is a learnable projection matrix and the attention coefficient $\alpha_{ij}^{(\ell)}$ is defined as:

$$\alpha_{ij}^{(\ell)} = \frac{\exp\Big(\text{LeakyReLU}\left(\mathbf{a}^\top[W^{(\ell)}\mathbf{h}_i^{(\ell-1)} \,||\, W^{(\ell)}\mathbf{h}_j^{(\ell-1)}]\right)\Big)}{\sum_{p\in\mathcal{N}(i)} \exp\Big(\text{LeakyReLU}\left(\mathbf{a}^\top[W^{(\ell)}\mathbf{h}_i^{(\ell-1)} \,||\, W^{(\ell)}\mathbf{h}_p^{(\ell-1)}]\right)\Big)}. \tag{7}$$

After L layers of attention-based message passing, we obtain graph-refined embeddings $G = [\mathbf{g}_1, \mathbf{g}_2, \ldots, \mathbf{g}_{T_{\text{OBS}}}] \in \mathbb{R}^{T_{\text{OBS}}\times D}$, the graph module captures semantic relationships between frames based on feature similarity; however, it does not explicitly encode temporal order. To model temporal ordering and long-range dependencies that are not captured by the graph structure, we further employ a Transformer encoder with positional embeddings, which provides global temporal modeling complementary to graph-based reasoning [22].

3.3 Temporal Encoding via Transformer Encoder

The details of the Transformer-based temporal encoding are described in the following section[1]. To capture global temporal structure, we use a Transformer encoder on the observed feature sequence $F_{\text{obs}} \in \mathbb{R}^{T_{\text{OBS}}\times D}$, Learnable positional embeddings $P \in \mathbb{R}^{T_{\text{OBS}}\times D}$ are added to preserve frame order and obtain:

$$Z_0 = F_{\text{obs}} + P. \tag{8}$$

The Transformer encoder consists of stacked self-attention layers that allow each frame to attend to all others, capturing long-range temporal dependencies such as early contextual cues, interaction buildup, and pre-action patterns. At each layer, multi-head self-attention is computed as

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V, \tag{9}$$

where, $Q = Z_0W_Q, \quad K = Z_0W_K, \quad V = Z_0W_V$, with $W_Q, W_K, W_V \in \mathbb{R}^{D\times D}$ and d denoting the per-head dimensionality. Following self-attention and feed-forward transformations with residual connections, the encoder produces a temporally contextualized representation

$$H = \text{Encoder}(Z_0) \in \mathbb{R}^{T_{\text{OBS}}\times D}. \tag{10}$$

The positional embeddings ensure that the encoder explicitly distinguishes early and late observations, enabling ordered temporal reasoning that complements similarity-based graph modeling.

Fusion of Graph and Temporal Reasoning. The GAT produces a relational representation $G \in \mathbb{R}^{T_{\text{OBS}}\times D}$, while the Transformer encoder outputs H, which encodes global temporal dynamics. To integrate these complementary cues, we concatenate and linearly project the two representations:

$$M = W_m[H \,||\, G] + b_m, \quad M \in \mathbb{R}^{T_{\text{OBS}}\times D}, \tag{11}$$

where, W_m and b_m are learnable parameters. The fused memory M jointly captures semantic relational structure and ordered temporal evolution, and serves as the input to the horizon-aware decoder for multi-step anticipation.

3.4 Horizon-Aware Transformer Decoder (HATD)

Anticipating actions at different future times requires modeling short-, mid-, and long-term cues. In egocentric videos, these cues range from immediate hand motion to broader object and scene context. To handle this, we introduce a Horizon-Aware Transformer Decoder (HATD) that uses learnable queries, each dedicated to a specific anticipation horizon, enabling time-aware future action prediction.

Learnable Horizon Queries: Instead of deriving queries from observed features, HATD defines a set of K learnable query vectors

$$Q = [\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_K] \in \mathbb{R}^{K \times D}, \tag{12}$$

where, each query $\mathbf{q}_k$ corresponds to the k-th anticipation horizon. These queries are optimized during training and act as horizon-specific information requests, encoding what type of future evidence should be extracted from the observed clip.

Cross-Attention over Graph-Enhanced Memory: Given the fused graph-enhanced temporal memory $M \in \mathbb{R}^{T_{\text{OBS}} \times D}$ produced by GETR, the decoder applies cross-attention between the horizon queries and the memory. The decoder in fact performs stacked cross-attention and feed-forward transformations, yielding horizon-specific latent representations

$$Y = \text{Decoder}(Q, M) = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_K] \in \mathbb{R}^{K \times D}. \tag{13}$$

Each representation $\mathbf{y}_k$ summarizes the observed sequence from the perspective of the corresponding anticipation horizon. Early queries naturally focus on recent motion cues, while distant queries emphasize higher-level context and long-term intent.

Horizon-Specific Classification Heads: For each horizon k, the final verb, noun, and action predictions are obtained using three independent linear classifiers:

$$\hat{\mathbf{v}}_k = W_v \mathbf{y}_k + b_v, \qquad \hat{\mathbf{n}}_k = W_n \mathbf{y}_k + b_n, \qquad \hat{\mathbf{a}}_k = W_a \mathbf{y}_k + b_a, \tag{14}$$

where, $\hat{\mathbf{v}}_k \in \mathbb{R}^{N_v}$, $\hat{\mathbf{n}}_k \in \mathbb{R}^{N_n}$, and $\hat{\mathbf{a}}_k \in \mathbb{R}^{N_a}$. This design enables explicit horizon-aware prediction without forcing a single representation to serve all future time scales.

Loss Functions: To effectively train our framework, we employ a combination of complementary losses that address prediction accuracy, class imbalance, temporal consistency, embedding discriminability, and graph structure regularization. The overall training objective is defined as the weighted sum of five loss terms:

$$\mathcal{L} = \mathcal{L}_{\mathcal{CE}} + \mathcal{L}_{\mathcal{F}} + \mathcal{L}_{\mathcal{TS}} + \mathcal{L}_{\mathcal{C}} + \mathcal{L}_{\mathcal{GR}}. \tag{15}$$

Here, $\mathcal{L}_{\mathcal{CE}}$ denotes the cross-entropy loss applied to horizon-wise verb, noun, and action predictions, $\mathcal{L}_{\mathcal{F}}$ the focal loss used to mitigate class imbalance, $\mathcal{L}_{\mathcal{TS}}$ enforces temporal smoothness across adjacent anticipation horizons, $\mathcal{L}_{\mathcal{C}}$ encourages discriminative horizon-specific embeddings through contrastive learning, and $\mathcal{L}_{\mathcal{GR}}$ regularizes the learned graph structure by aligning it with the original k-nearest-neighbor similarity graph. Each loss component is defined below:

Cross-Entropy Loss. Given the decoder outputs $\hat{\mathbf{y}}_k$ at anticipation horizon $k \in \{1, \ldots, K\}$ and the corresponding ground-truth labels y_k, we apply a cross-entropy loss

$$\mathcal{L}_{\mathcal{CE}} = -\sum_{k=1}^{K} (\mathbf{y}_k \neq -1) \log p(\hat{\mathbf{y}}_k = \mathbf{y}_k), \tag{16}$$

Focal Loss. To address severe class imbalance in egocentric action anticipation, we further apply focal loss

$$\mathcal{L}_{\mathcal{F}} = -\sum_{k=1}^{K} (1 - p_k)^{\gamma} \log p_k, \tag{17}$$

where, p_k denotes the predicted probability of the ground-truth class at horizon k, and γ controls the focusing strength.

Temporal Smoothness Loss. Since adjacent anticipation horizons correspond to closely related future states, we enforce temporal consistency by minimizing which penalizes abrupt prediction changes across horizons.

$$\mathcal{L}_{\mathcal{TS}} = \frac{1}{K-1} \sum_{k=1}^{K-1} \left\| \hat{\mathbf{y}}_{k+1} - \hat{\mathbf{y}}_k \right\|_2^2, \tag{18}$$

Contrastive Loss. To enhance discriminability of horizon-aware representations, we introduce a contrastive loss on decoder embeddings $\mathbf{y}_k$:

$$\mathcal{L}_{\mathcal{C}} = \sum_{(i,j)} \left[(y_i = y_j) \|\mathbf{y}_i - \mathbf{y}_j\|_2^2 + (y_i \neq y_j) \max(0, m - \|\mathbf{y}_i - \mathbf{y}_j\|_2)^2 \right], \tag{19}$$

where, m is a margin.

Graph Reconstruction Loss. Finally, to regularize the graph attention module, we reconstruct the similarity structure among observed frames. Let A denote the original k-NN adjacency matrix and $\hat{A}$ the reconstructed one from GAT embeddings. The loss is defined as

$$\mathcal{L}_{\mathcal{GR}} = \left\| A - \hat{A} \right\|_F^2. \tag{20}$$

4 Experimental Results

In this section, we evaluate our approach on two large-scale egocentric action datasets, EPIC-Kitchens [4] and EGTEA Gaze+ [13]. All experiments were conducted on a workstation equipped with an Intel Xeon W-2295 CPU (18 cores at 3.00 GHz), 128 GB DDR4 RAM, and an NVIDIA RTX A4000 GPU.

4.1 Implementation Details

Our framework is implemented in PyTorch and PyTorch Geometric using a unified pipeline across all datasets. Training samples use a 90-frame observation window that ends in each annotated endframe. Future supervision is defined time-wise for anticipation horizons from 0.25–2.0s, we locate the corresponding future frame and assign verb, noun, and action labels. The model integrates a k-NN based graph encoder with a three-layer GAT and a three-layer Transformer encoder, followed by a Transformer decoder driven by learned horizon queries. All training hyperparameters are reported in Table 1.

Table 1: Hyperparameter Details of the Proposed Framework

Hyperparameter	**Value**
Dropout	0.1
Batch Size	8
Optimizer	Adam
Learning Rate (LR)	1×10^{-4}
Weight Decay (WD)	1×10^{-4}
Training Epochs	100

4.2 Comparison with State-of-the-Art Methods

Results on the EPIC-Kitchens Dataset: We compare our proposed Ego-HAnG framework with seven state-of-the-art methods 2SCNN [4], RULSTM [7], LAEAI [23], RAFTformer-2B [9], InAViT [19], PlausiVL [16] and MF+ActSeq [17] on the EPIC-Kitchens dataset [4], as summarized in Table 2. The Table shows that the proposed method consistently outperforms the second-best approach, across all evaluated settings. Specifically, it achieves higher Top-1 and Top-5 accuracy for verb, noun, and action anticipation, along with improved Top-5 action recall.

Table 2: Action Anticipation Performance on EPIC-KITCHENS dataset

Methods	**Top-1 Accuracy**			**Top-5 Accuracy**			**Top-5 Recall**
	Verb	**Noun**	**Action**	**Verb**	**Noun**	**Action**	**Action**
2SCNN (ECCV'2018) [4]	29.76	15.15	04.32	76.03	38.76	15.21	–
RULSTM (ICCV'2019) [7]	33.04	22.78	14.39	79.55	50.95	33.73	13.30
LAEAI (TIP'2020) [23]	35.44	22.79	14.66	79.72	50.09	34.98	–
RAFTformer-2B (CVPR'2023) [9]	–	–	–	–	–	–	19.10
InAViT (WACV'2024) [19]	49.14	49.97	23.75	–	–	–	25.89
PlausiVL (CVPR'2024) [16]	–	–	–	–	–	–	27.60
MF+ActSeq (ICLR'2025) [17]	–	–	–	–	–	–	22.40
EgoHAnG (Ours)	**50.85**	**50.72**	**24.40**	**80.02**	**52.99**	**35.64**	**28.71**
Improv. over 2nd best	+1.71	+0.75	+0.65	+0.10	+2.04	+0.66	+1.11

In addition, we show a detailed evaluation under varying anticipation horizons in Table 3, which reports Top-5 action accuracy across eight anticipation times ranging from 2.0 to 0.25 s. The four methods *i.e.*, FN [5], RL [15], and RULSTM [7] all anticipation times and one method *i.e.*, LAEAI [23] provides results from 1.50 s to 0.25 s. Across all horizons, our proposed EgoHAnG consistently outperforms existing approaches and achieves clear improvements over the second-best method (show in underline), demonstrating its robustness across both long-term and short-term anticipation settings. To further complement the quantitative evaluation, we present **qualitative results** of our action anticipation model on the EPIC-Kitchens dataset in Fig. 3. From right to left, the observation window gradually moves closer to the future action. The orange lighted frame indicates the action of ground-truth, while the Top-5 predictions of the model are shown for anticipation horizons $T \in \{0.5s, 1.0s, 1.5s, 2.0s\}$. Correct predictions are highlighted in green. In the first example (*wash celery*), the model correctly predicts the action at all anticipation times, demonstrating stable performance even at longer horizons.

Table 3: Action Accuracy across Time Horizons on EPIC-KITCHENS dataset.

Methods	**Top-5 Action Accuracy (%) at Different Time (τ_a)**							
	2.0	1.75	1.50	1.25	1	0.75	0.50	0.25
RL (CVPR'2016) [15]	25.95	26.49	27.15	28.48	29.61	30.81	31.86	32.84
FN (WACV'2018) [5]	23.47	24.07	24.68	25.66	26.27	26.87	27.88	28.96
RULSTM (ICCV'2019) [7]	29.44	30.73	32.24	33.41	35.32	36.34	37.37	38.98
LAEAI (TIP'2020) [23]	–	–	32.50	33.60	35.60	36.70	38.50	39.40
EgoHAnG (Ours)	**33.23**	**33.63**	**34.51**	**33.62**	**35.64**	**37.11**	**38.71**	**39.92**
Improv. over 2nd best	+3.79	+2.90	+2.01	+0.02	+0.04	+0.41	+0.21	+0.52

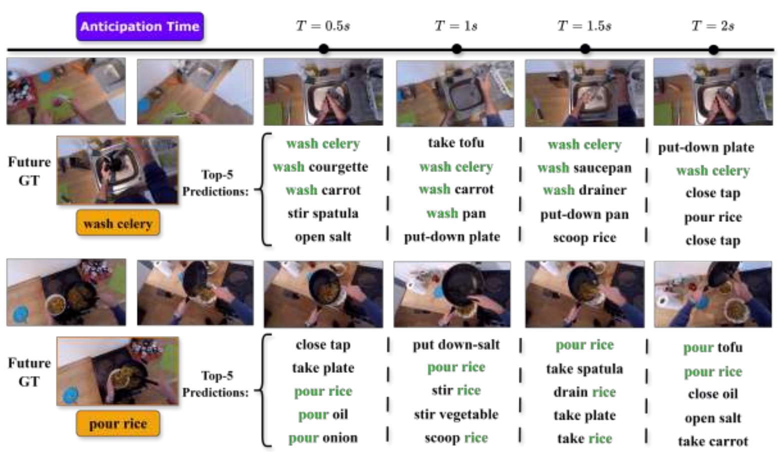

Fig. 3: Qualitative results at different anticipation horizons ($T = \{0.5, 1.0, 1.5, 2.0\}s$). Top-5 predictions are shown; orange marks the ground truth and green indicates (ours) correct anticipations.

Results on the EGTEA Gaze+ Dataset: We compare our proposed EgoHAnG framework with six state-of-the-art methods, namely FHOI [14], AVT [10], AFFT [24], Abs. Goal [18], InAViT [19], and the recent RAFTformer+ActSeq (ICLR'2025) [17], on the EGTEA Gaze+ dataset [13], as shown in Table 4. As observed, EgoHAnG achieves the best performance across all settings, clearly outperforming the second-best approach. In particular, it improves Top-1 accuracy by +9.92%, +5.07%, and +3.60% for verb, noun, and action prediction, respectively, and achieves a +3.73% gain in Top-5 action recall, demonstrating strong generalization on EGTEA Gaze+ dataset.

Table 4: Action Anticipation Performance on EGTEA Gaze+ dataset.

Methods	**Top-1 Accuracy**			**Top-5 Recall**
	Verb	**Noun**	**Action**	**Action**
FHOI (ECCV'2020) [14]	49.00	45.50	36.60	36.40
AVT (ICCV'2021) [10]	54.90	52.20	43.00	35.20
AFFT (WACV'2023) [24]	53.40	50.40	42.50	35.20
Abs. Goal (ICPR'2024) [18]	64.80	65.30	49.80	37.40
InAViT (WACV'2024) [19]	<u>79.30</u>	<u>77.60</u>	<u>67.80</u>	58.20
RAFTformer+ActSeq (ICLR'2025) [17]	–	–	–	<u>66.30</u>
EgoHAnG (Ours)	**89.22**	**82.67**	**71.40**	**70.03**
Improv. over 2nd best	+9.92	+5.07	+3.60	+3.73

Table 5 summarizes the Top-5 action anticipation accuracy on the EGTEA Gaze+ dataset at multiple anticipation horizons. The proposed EgoHAnG delivers the strongest results at all time points and consistently surpasses the second-best approach. The advantage is most pronounced at earlier anticipation stages, achieving improvements of +10.06%, +7.90%, and +6.48% at 2.0 s, 1.75 s, and 1.50 s, respectively, highlighting the effectiveness of our model for long-range anticipation while remaining robust at shorter horizons.

Table 5: Action Accuracy across Time Horizons on EGTEA Gaze+ dataset.

Methods	**Top-5 Action Accuracy (%) at Different Time (τ_a)**							
	2.0	1.75	1.50	1.25	1	0.75	0.50	0.25
FN (WACV'2018) [5]	54.06	54.94	56.75	58.34	60.12	62.03	63.93	66.45
RL (CVPR'2016) [15]	55.18	56.31	58.22	60.35	62.56	64.65	67.35	70.42
RULSTM (ICCV'2019) [7]	<u>56.82</u>	<u>59.13</u>	<u>61.42</u>	<u>63.53</u>	66.40	68.41	71.84	74.28
LAEAI (TIP'2020) [23]	–	–	–	–	<u>66.71</u>	<u>68.54</u>	<u>72.32</u>	<u>74.59</u>
EgoHAnG (Ours)	**66.88**	**67.03**	**67.90**	**68.22**	**70.17**	**71.61**	**72.40**	**74.88**
Improv. over 2nd best	+10.06	+7.90	+6.48	+4.69	+3.46	+3.07	+0.08	+0.29

4.3 Ablation studies

To demonstrate successive improvements, the following ablation studies are carried out, impact of the use of PCA, different values of k for k-NN graph construction, the impact of removing GETR or HATD, and the effect of different loss functions.

Impact of using PCA: Table 6 shows the effect of using PCA on feature representations. Compared to InAViT [19] and our model without PCA, applying PCA greatly lowers computation (150 GFlops vs. 600/391 GFlops) and model parameter (26.3M vs. 398.1M/157.2M) while improving Top-5 action accuracy at all anticipation times, showing that PCA makes the model faster and more accurate.

Table 6: Impact of PCA on EGTEA Gaze+ dataset.

Methods	Anticipation Gap τ_a (s)				GFlops	Params
	2.0	1.5	1.0	0.5		
InAViT [19]	64.10	65.80	66.90	67.80	391	157.2M
w/o PCA	64.67	66.20	66.98	67.44	600	398.1M
with PCA	66.88	67.90	70.17	71.40	150	26.3M

Impact of k-NN Graph In the next ablation study, we analyze the impact of neighborhood size k. Table 7 reports the effect of different k values in the k-NN graph and modality choices on the accuracy of Top-5 and the mean recall of Top-5 in EPIC-Kitchens. For both RGB and optical flow, $k = 5$ gives the best performance, while smaller ($k = 3$) or larger ($k = 10$) values slightly degrade results. Using RGB and optical flow together consistently outperforms single modalities, showing the benefit of multimodal fusion and appropriate graph connectivity.

Table 7: Impact of k and Modality on Top-5 Accuracy and Top-5 Action Recall.

Modality	**Different** K Value	**Top-5 Accuracy**			**Top-5 Recall**
		Verb	**Noun**	**Action**	**Action**
RGB	$k = 3$	70.82	43.11	26.72	20.65
	$k = 5$	72.94	45.33	28.48	19.82
	$k = 10$	69.05	41.72	26.01	18.90
Optical Flow	$k = 3$	62.14	35.03	19.88	16.42
	$k = 5$	63.26	36.21	20.55	17.01
	$k = 10$	61.09	33.87	19.21	15.63
RGB + Optical Flow	$k = 3$	79.45	52.28	34.94	24.04
	$k = 5$	**80.02**	**52.99**	**35.64**	**28.71**
	$k = 10$	76.20	49.17	34.36	21.09

Impact of different modules (GETR and HATD): Table 8 shows Top-1 accuracy for verb, noun, and action using RGB + optical flow. Compared to the baseline InAViT [19], the full EgoHAnG model achieves the highest accuracy in all categories. Removing the GETR module reduces the model's ability to capture temporal relationships, while removing HATD affects time-aware predictions. These results show that both modules are important for improving overall performance.

Effects of Different Components of the Loss Function: Table [4] reports Top-5 accuracy (verb, noun, action) and Top-5 action recall on EPIC-Kitchens under different loss combinations. The baseline cross-entropy loss ($\mathcal{L}_{\mathcal{CE}}$) shows limited performance, especially for actions. Focal loss ($\mathcal{L}_{\mathcal{F}}$) improves results by addressing class imbalance, while temporal smoothness ($\mathcal{L}_{\mathcal{TS}}$) increases recall by enforcing consistency across horizons. Contrastive loss ($\mathcal{L}_{\mathcal{C}}$) further boosts noun and action accuracy, and graph reconstruction ($\mathcal{L}_{GR}$) provides the strongest single-loss gain by enhancing relational learning. Combining all losses yields the best performance, achieving 80.02%, 52.99%, and 35.64% Top-5 accuracy for verb, noun, and action, and 28.71% action recall.

Table 8: Impact of GETR and HATD on Top-1 Accuracy

Modality	**Methods**	**Top-1 Accuracy**		
		Verb	**Noun**	**Action**
RGB + OpticalFlow	InAViT [19]	49.14	49.97	23.75
	Ours w/o GETR	49.21	49.67	22.17
	Ours w/o HATD	48.94	49.08	23.81
	Ours	**50.85**	**50.72**	**24.40**

Table 9: Impact of Individual Loss function on Action Anticipation Performance.

Loss	**Top-5 Accuracy**			**Top-5 Recall**
	Verb	**Noun**	**Action**	**Action**
$\mathcal{L}_{\mathcal{CE}}$	50.17	24.78	10.34	12.77
$\mathcal{L}_{\mathcal{CE}}$ + $\mathcal{L}_{\mathcal{F}}$	51.65	25.33	13.09	14.02
$\mathcal{L}_{\mathcal{CE}}$ + $\mathcal{L}_{\mathcal{TS}}$	50.87	20.27	12.17	13.19
$\mathcal{L}_{\mathcal{CE}}$ + $\mathcal{L}_{\mathcal{C}}$	54.34	27.11	13.88	14.73
$\mathcal{L}_{\mathcal{CE}}$ + $\mathcal{L}_{\mathcal{GR}}$	55.86	30.66	17.54	17.98
$\mathcal{L}_{\mathcal{CE}}$ + $\mathcal{L}_{\mathcal{F}}$ + $\mathcal{L}_{\mathcal{TS}}$	65.33	41.35	25.56	22.21
$\mathcal{L}_{\mathcal{CE}}$ + $\mathcal{L}_{\mathcal{F}}$ + $\mathcal{L}_{\mathcal{C}}$	72.78	44.14	27.67	23.09
$\mathcal{L}_{\mathcal{CE}}$ + $\mathcal{L}_{\mathcal{F}}$ + $\mathcal{L}_{\mathcal{GR}}$	69.12	43.91	31.97	24.66
$\mathcal{L}_{\mathcal{CE}}$+$\mathcal{L}_{\mathcal{F}}$ +$\mathcal{L}_{\mathcal{TS}}$+$\mathcal{L}_{\mathcal{C}}$+$\mathcal{L}_{\mathcal{GR}}$ **(Ours)**	**80.02**	**52.99**	**35.64**	**28.71**

5 Conclusion

In this paper, we proposed a novel framework, EgoHAnG, for egocentric action anticipation. Our approach combines early RGB–flow fusion, graph-based temporal reasoning, and a horizon-aware Transformer decoder to effectively model inter-frame relationships and long-range temporal dependencies. Extensive experiments on EPIC-Kitchens and EGTEA Gaze+ datasets demonstrate that our method outperforms state-of-the-art approaches across multiple metrics and anticipation horizons. The results highlight the effectiveness and robustness of integrating relational structure learning with multimodal temporal modeling for anticipating future actions in egocentric videos. In future, we will extend our framework to anticipate multiple concurrent actions.

References

1. Aitken, K., Ramasesh, V., Cao, Y., Maheswaranathan, N.: Understanding how encoder-decoder architectures attend. NeurIPS **34**, 22184–22195 (2021)
2. Akilan, T., Wu, Q.J., Safaei, A., Huo, J., Yang, Y.: A 3d cnn-lstm-based image-to-image foreground segmentation. IEEE Trans. Intell. Transp. Syst. **21**(3), 959–971 (2019)

3. Chowdhury, A.S., Sahu, A.: Graph Based Multimedia Analysis. Elsevier (2024)
4. Damen, D., Doughty, H., Farinella, G.M., Fidler, S., Furnari, A., Kazakos, E., Moltisanti, D., Munro, J., Perrett, T., Price, W., et al.: Scaling egocentric vision: The epic-kitchens dataset. In: Proc. ECCV. pp. 720–736 (2018)
5. De Geest, R., Tuytelaars, T.: Modeling temporal structure with lstm for online action detection. In: Proc. WACV. pp. 1549–1557. IEEE (2018)
6. Feichtenhofer, C.: X3d: Expanding architectures for efficient video recognition. In: Proc. CVPR. pp. 203–213 (2020)
7. Furnari, A., Farinella, G.M.: What would you expect? anticipating egocentric actions with rolling-unrolling lstms and modality attention. In: Proc. ICCV. pp. 6252–6261 (2019)
8. Gao, R., Oh, T.H., Grauman, K., Torresani, L.: Listen to look: Action recognition by previewing audio. In: Proc. CVPR. pp. 10457–10467 (2020)
9. Girase, H., Agarwal, N., Choi, C., Mangalam, K.: Latency matters: Real-time action forecasting transformer. In: In Proc. CVPR. pp. 18759–18769 (2023)
10. Girdhar, R., Grauman, K.: Anticipative video transformer. In: Proc. ICCV. pp. 13505–13515 (2021)
11. Gong, D., Lee, J., Kim, M., Ha, S.J., Cho, M.: Future transformer for long-term action anticipation. In: Proc. CVPR. pp. 3052–3061 (2022)
12. Li, H., Zheng, W.S., Zhang, J., Hu, H., Lu, J., Lai, J.H.: Egocentric action recognition by automatic relation modeling. IEEE Trans. Pattern Anal. Mach. Intell. **45**(1), 489–507 (2022)
13. Li, Y., Liu, M., Rehg, J.M.: In the eye of the beholder: Gaze and actions in first person video. IEEE Trans. Pattern Anal. Mach. Intell. **45**(6), 6731–6747 (2021)
14. Liu, M., Tang, S., Li, Y., Rehg, J.M.: Forecasting human-object interaction: joint prediction of motor attention and actions in first person video. In: Proc. ECCV. pp. 704–721. Springer (2020)
15. Ma, S., Sigal, L., Sclaroff, S.: Learning activity progression in lstms for activity detection and early detection. In: Proc. CVPR. pp. 1942–1950 (2016)
16. Mittal, H., Agarwal, N., Lo, S.Y., Lee, K.: Can't make an omelette without breaking some eggs: Plausible action anticipation using large video-language models. In: Proc. CVPR. pp. 18580–18590 (2024)
17. Qiu, Y., Rajan, D.: Action sequence augmentation for action anticipation. In: Proc. ICLR. pp. 1–18 (2025)
18. Roy, D., Fernando, B.: Predicting the next action by modeling the abstract goal. In: Proc. ICPR. pp. 162–177. Springer (2024)
19. Roy, D., Rajendiran, R., Fernando, B.: Interaction region visual transformer for egocentric action anticipation. In: Proc. WACV. pp. 6740–6750 (2024)
20. Sahu, A., Chowdhury, A.S.: Together recognizing, localizing and summarizing actions in egocentric videos. IEEE Trans. Image Process. **30**, 4330–4340 (2021)
21. Shaikh, M.B., Chai, D., Islam, S.M.S., Akhtar, N.: From cnns to transformers in multimodal human action recognition: A survey. ACM Trans. Multimed. Comput. Commun. Appl. **20**(8), 1–24 (2024)
22. Veličković, P., Cucurull, G., Casanova, A., Romero, A., Lio, P., Bengio, Y.: Graph attention networks. arXiv preprint arXiv:1710.10903 (2017)
23. Wu, Y., Zhu, L., Wang, X., Yang, Y., Wu, F.: Learning to anticipate egocentric actions by imagination. IEEE Trans. Image Process. **30**, 1143–1152 (2020)
24. Zhong, Z., Schneider, D., Voit, M., Stiefelhagen, R., Beyerer, J.: Anticipative feature fusion transformer for multi-modal action anticipation. In: Proc. WACV. pp. 6068–6077 (2023)